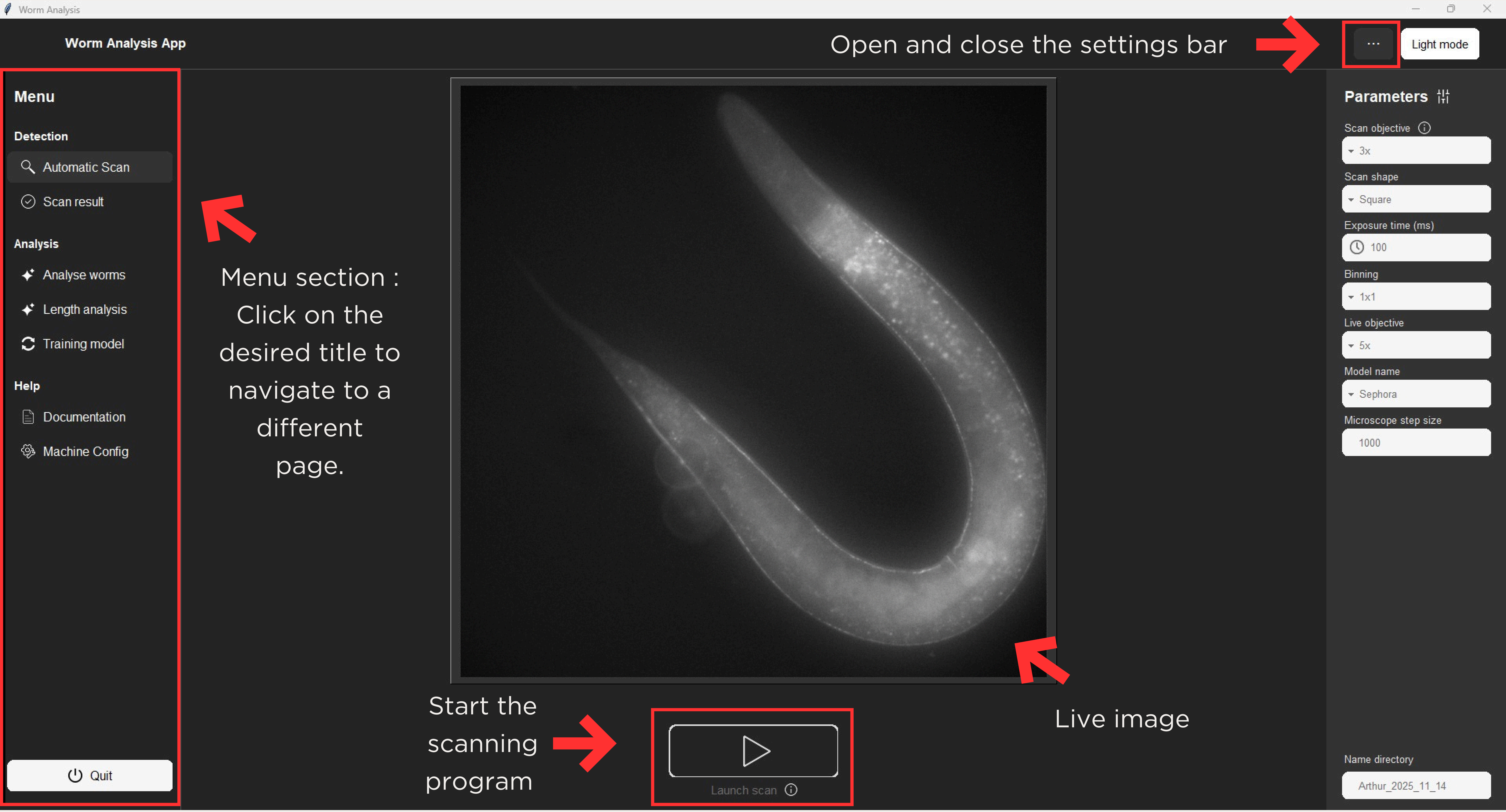
USER GUIDE
FOR THE SCANNING AND ANALYSIS SOFTWARE

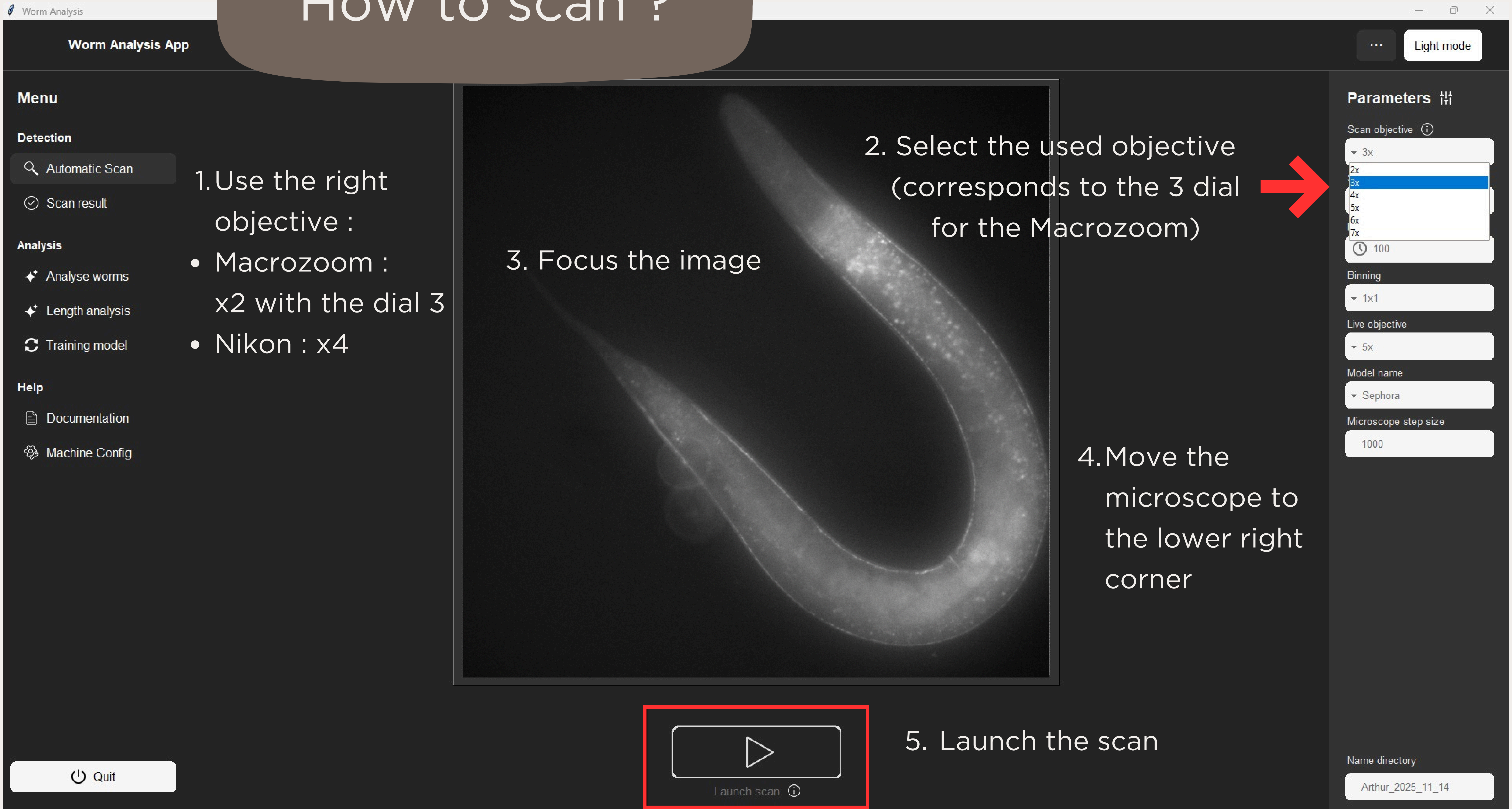
ARTHUR SKOWRONEK

arthur.skowronek13@gmail.com


Worm Analysis
Worm Analysis App
Open and close the settings bar
Light mode
Menu
Detection
Automatic Scan
Scan result
Analysis
Analyse worms
Length analysis
Training model
Help
Documentation
Machine Config
Quit
Menu section : Click on the desired title to navigate to a different page.
Parameters
Scan objective
3x
Scan shape
Square
Exposure time (ms)
100
Binning
1x1
Live objective
5x
Model name
Sephora
Microscope step size
1000
Name directory
Arthur_2025_11_14
Start the scanning program
Launch scan
Live image

How to scan ?

1. Use the right objective :
   - Macrozoom : x2 with the dial 3
   - Nikon : x4
2. Select the used objective (corresponds to the 3 dial for the Macrozoom)
3. Focus the image
4. Move the microscope to the lower right corner
5. Launch the scan



NOTES

1. If you are looking at tiny worms : you can use a higher magnification to do the scan.

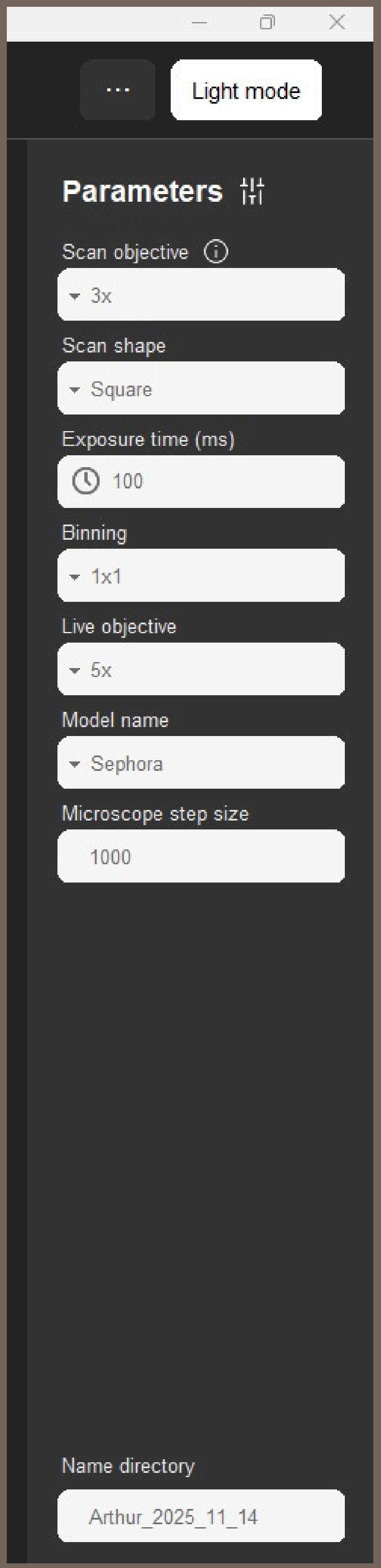
<u>For the Macrozoom :</u> ALWAYS use the x2 objective. But you can change the dial.

<u>For the Nikon :</u> Use whichever objective you prefer

<u>In both cases :</u> Don’t forget to update the value in the parameters panel in “Scan Objective”

2. You can set the scan shape to either square or rectangle.

Scan shape :



Worm Analysis

Worm Analysis App

··· Light mode

Menu

Detection

- Automatic Scan
- Scan result

Analysis

- Analyse worms
- Length analysis
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- Documentation
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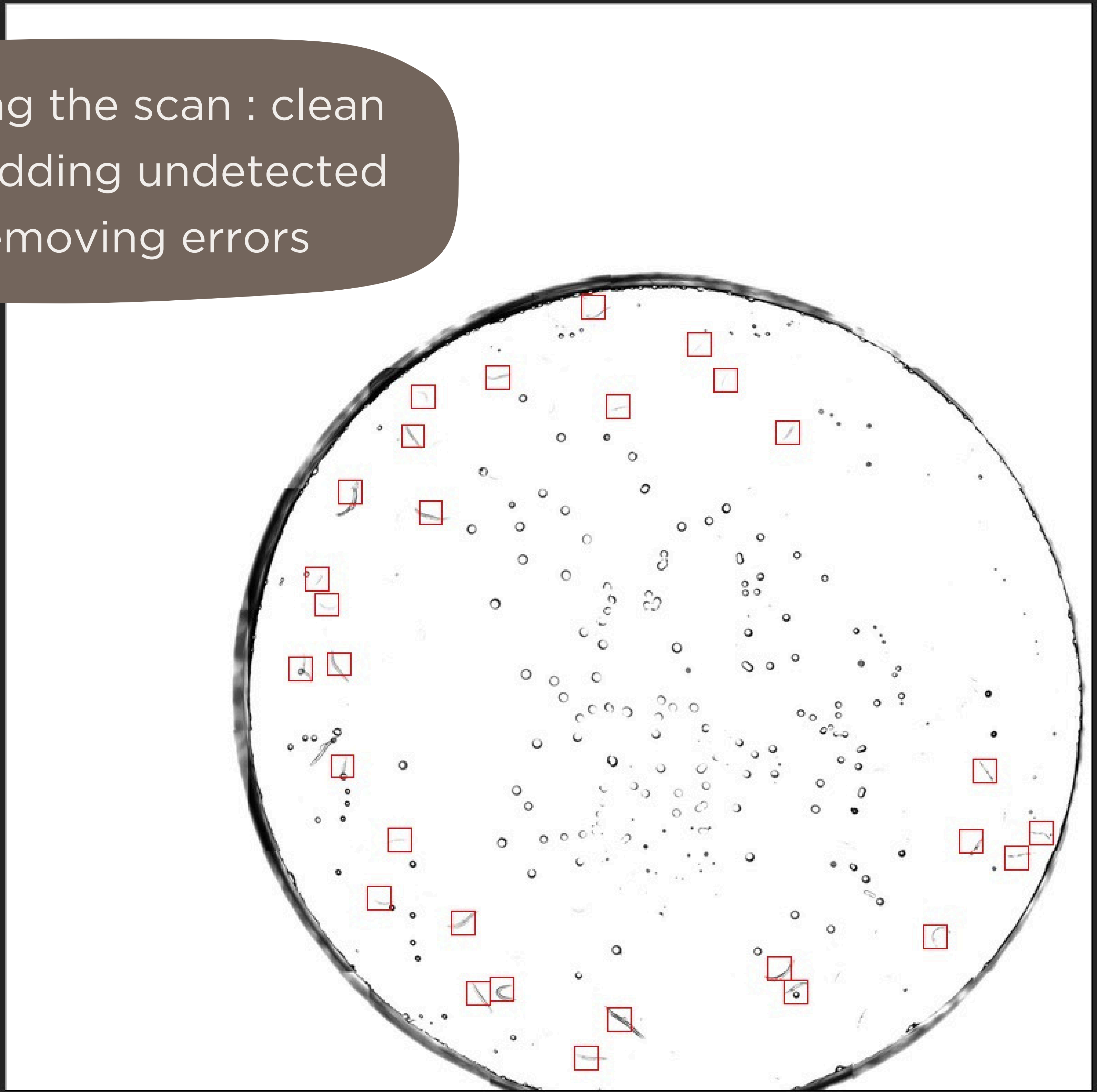
After performing the scan : clean the results by adding undetected worms or removing errors



- **Add worm :** you only have to click on the image where you want to add a worm
- **Remove worm :** click on the worm you want to delete. Or drag your mouse on the worms
- **Move worm :** click on a red rectangle and drag it to the desired position

Select the mode you want to use to clean the worm detection

Parameters

Scan objective
3x

Scan shape
Square

Exposure time (ms)
100

Binning
1x1

Live objective
5x

Model name
Sephora

Microscope step size
1000

Name directory
Arthur_2025_11_14

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Live
Snap image
Save image
Counter : Displays the ID of the worm being viewed over the total number of detected worms
1/29
Next/Last worm
Use the space bar to move to the next worm.
Tell the microscope to move to the next (or last) worm position
You can save an image regardless of the mode used (live or snap). The image will be saved in the folder defined here:
(the folder will be created automatically if it does not exist)
Parameters
Scan objective
3x
Scan shape
Square
Exposure time (ms)
100
Binning
1x1
Live objective
5x
Model name
Sephora
Microscope step size
1000
Name directory
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Live
Snap image
Save image
You can tag the worm being viewed as having a mutation or not
Wild-Type
100%
Mutation
0%
By hovering the info icon you can see the list of the ID of the worm being tag as Mutant
Next/Last mutant
Tell the microscope to move to the next (or last) worm with a mutation
Parameters
Scan objective
3x
Scan shape
Square
Exposure time (ms)
100
Binning
1x1
Live objective
5x
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Sephora
Microscope step size
1000
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Live
Snap image
Save image
Pixel Intensity Histogram
800000
600000
400000
200000
0
340
360
380
400
420
440
460
480
vmin
vmax
387
415
Auto
Full
Change the image displayed with the sliders and buttons
Parameters
Scan objective
3x
Scan shape
Square
Exposure time (ms)
100
Binning
1x1
Live objective
5x
Model name
Sephora
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1000
Name directory
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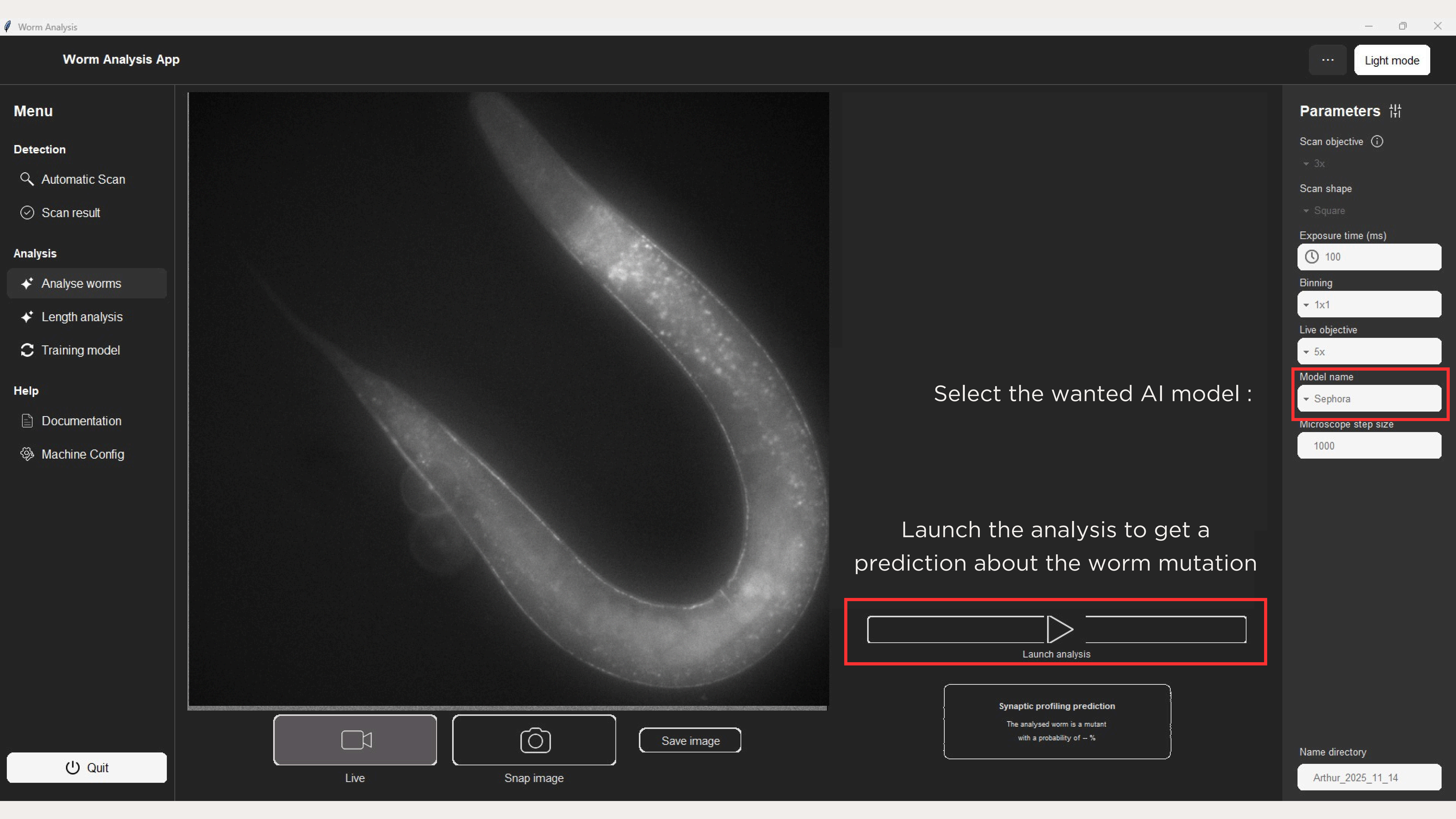
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Live
Snap image
Save image
Select the wanted AI model :
Launch the analysis to get a prediction about the worm mutation
Launch analysis
Synaptic profiling prediction
The analysed worm is a mutant with a probability of -- %
Parameters
Scan objective
3x
Scan shape
Square
Exposure time (ms)
100
Binning
1x1
Live objective
5x
Model name
Sephora
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1000
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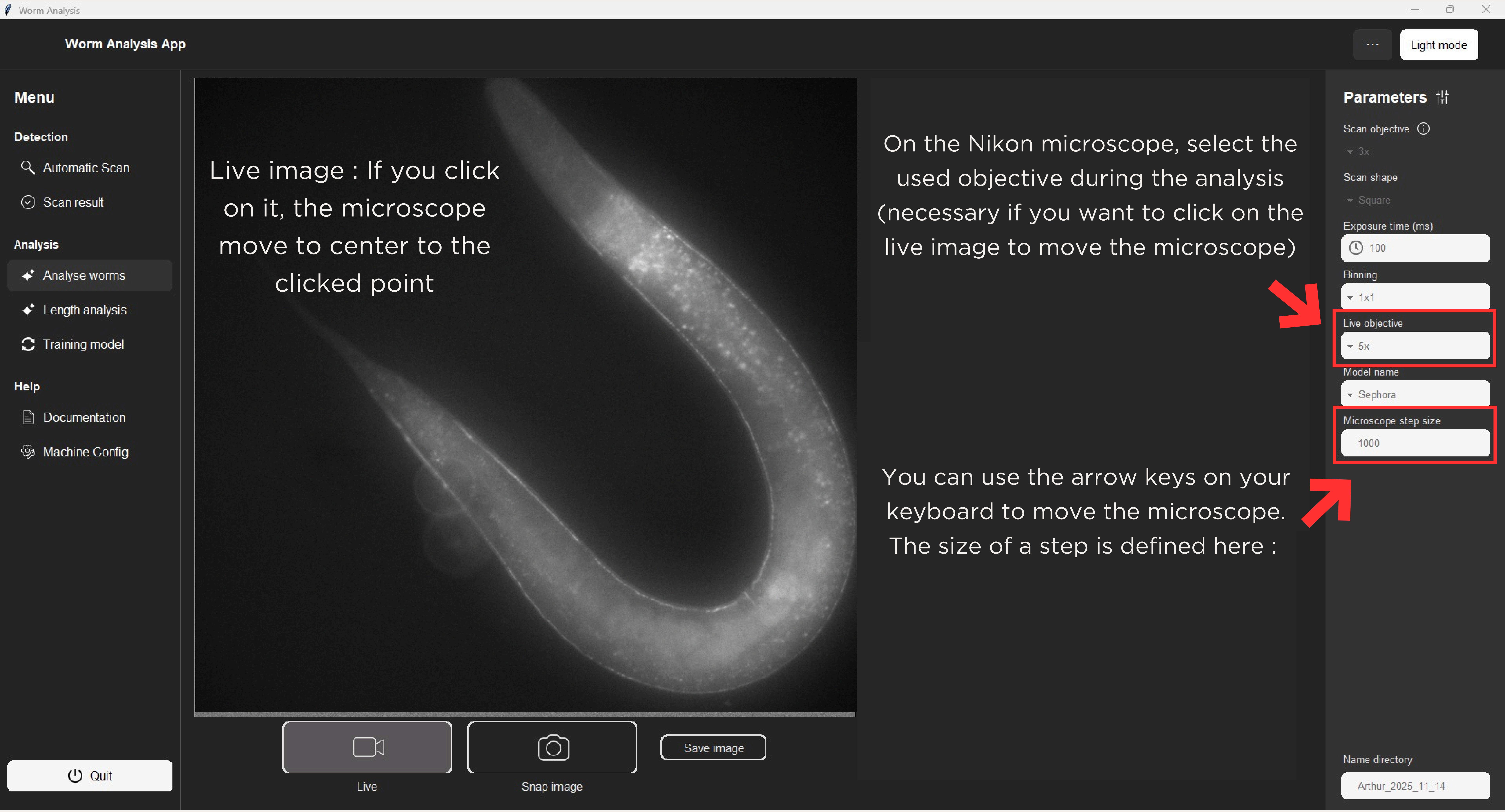
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Live image : If you click on it, the microscope move to center to the clicked point
Live
Snap image
Save image
On the Nikon microscope, select the used objective during the analysis (necessary if you want to click on the live image to move the microscope)
You can use the arrow keys on your keyboard to move the microscope. The size of a step is defined here :
Parameters
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3x
Scan shape
Square
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100
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1x1
Live objective
5x
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0
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360
380
400
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vmin
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387
415
Auto
Full
Wild-Type
100%
Mutation
0%
1/29
Next/Last worm
Use the space bar to move to the next worm.
Next/Last mutant
Launch analysis
Synaptic profiling prediction
The analysed worm is a mutant
with a probability of -- %
Parameters
Scan objective
3x
Scan shape
Square
Exposure time (ms)
100
Binning
1x1
Live objective
5x
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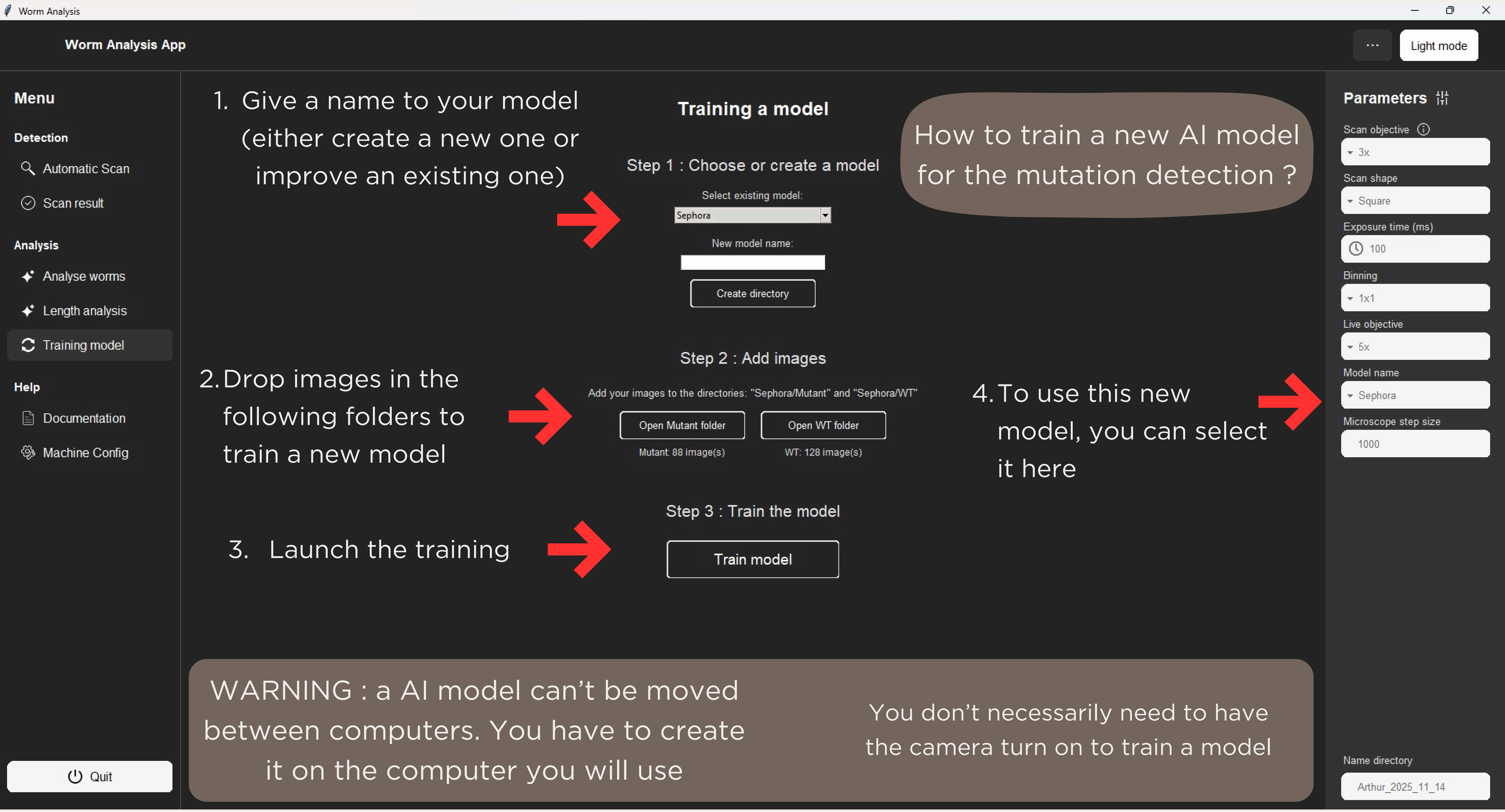
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1. Give a name to your model (either create a new one or improve an existing one)
Training a model
Step 1 : Choose or create a model
Select existing model:
Sephora
New model name:
Create directory
How to train a new AI model for the mutation detection ?
Step 2 : Add images
2. Drop images in the following folders to train a new model
Add your images to the directories: "Sephora/Mutant" and "Sephora/WT"
Open Mutant folder
Open WT folder
Mutant: 88 image(s)
WT: 128 image(s)
4. To use this new model, you can select it here
Step 3 : Train the model
3. Launch the training
Train model
WARNING : a AI model can't be moved between computers. You have to create it on the computer you will use
You don't necessarily need to have the camera turn on to train a model
Parameters
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3x
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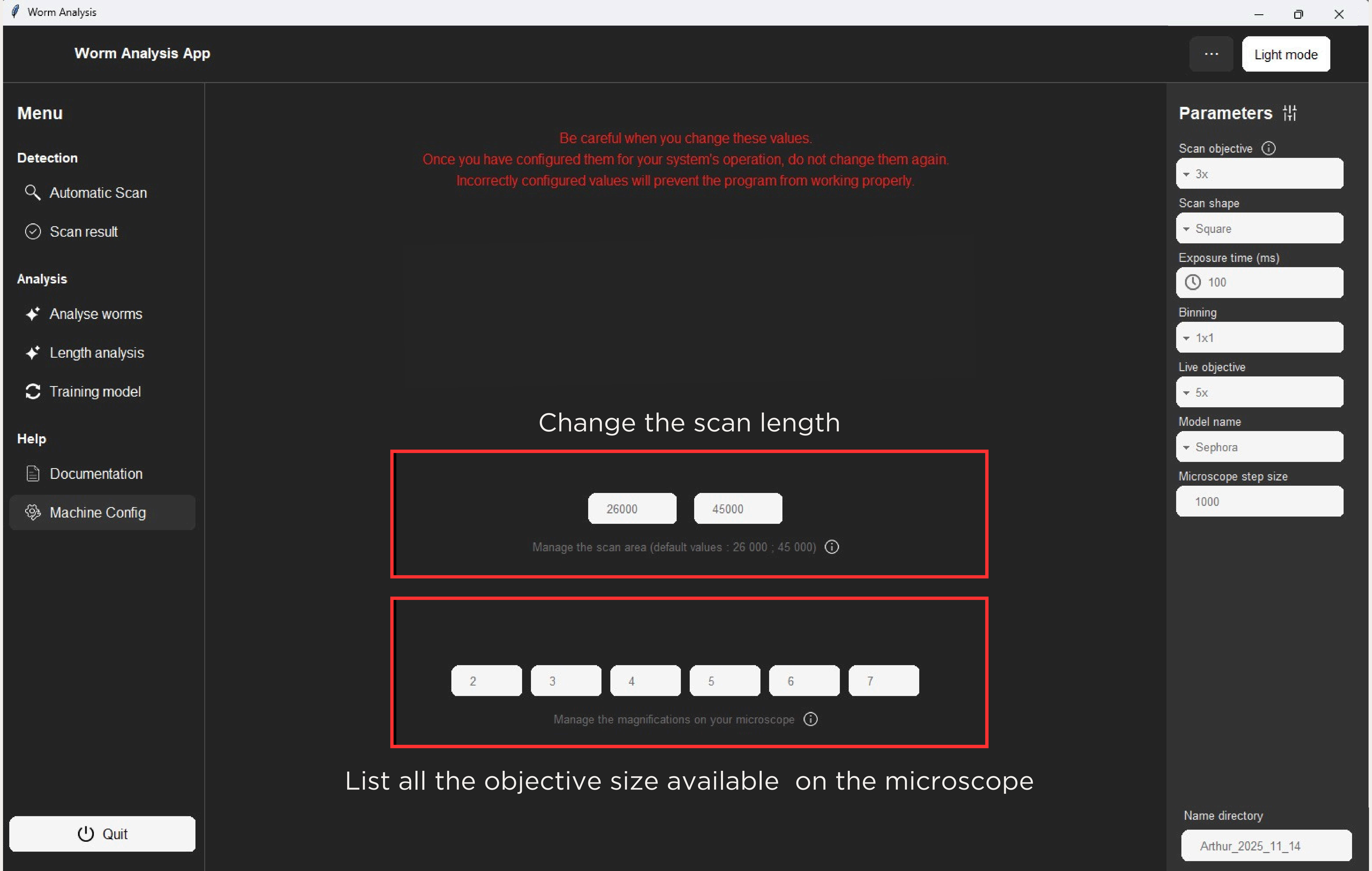
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Be careful when you change these values.
Once you have configured them for your system's operation, do not change them again.
Incorrectly configured values will prevent the program from working properly.
Change the scan length
26000
45000
Manage the scan area (default values : 26 000 ; 45 000)
2
3
4
5
6
7
Manage the magnifications on your microscope
List all the objective size available on the microscope
Parameters
Scan objective
3x
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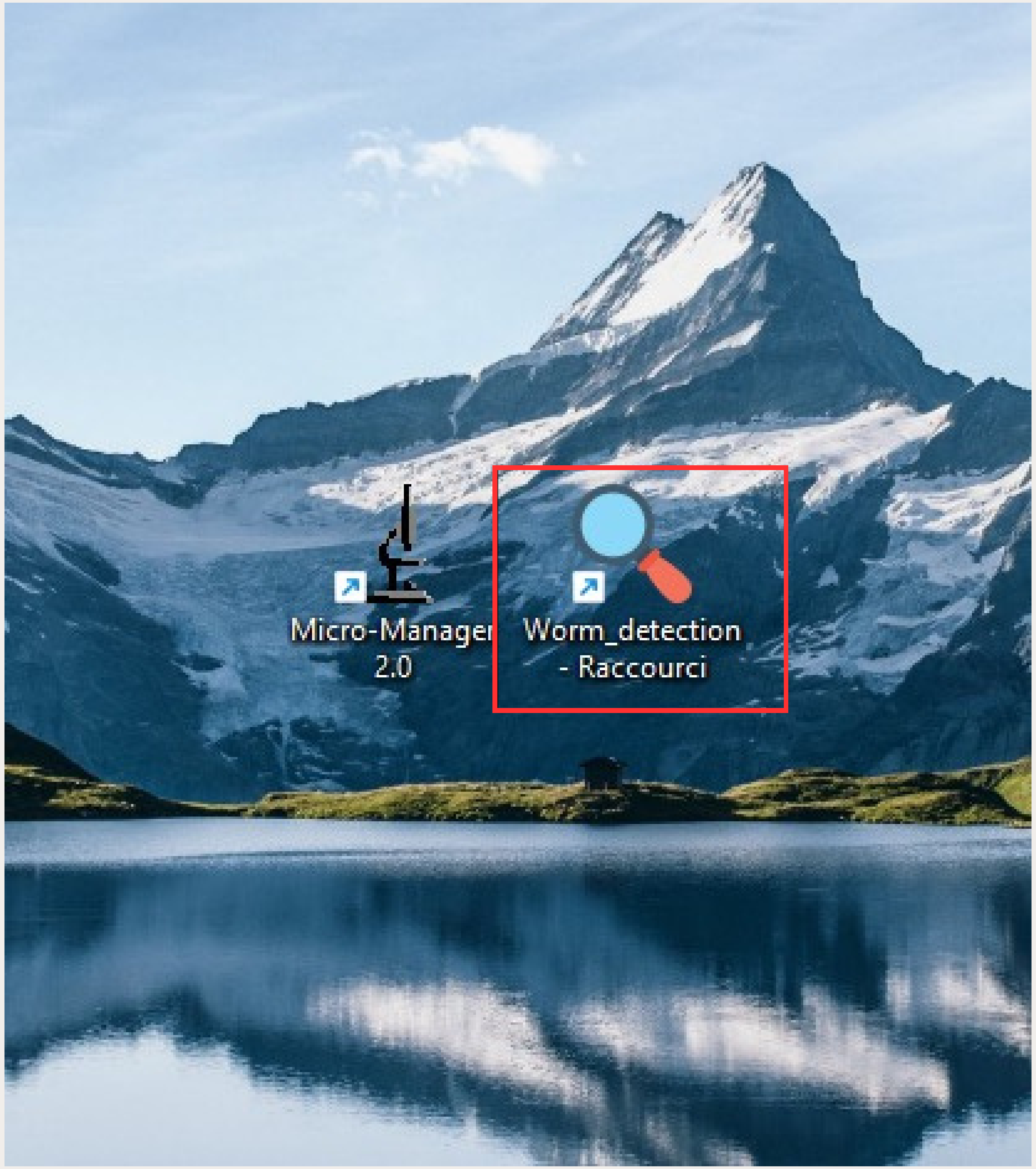

Click on this app to open the software.
The launching can take 10-20s.